



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 • STANDARD 6.AT.2

Understand Rates and Unit Rates

Lesson 3-2 • Teacher Copy

Name: _____

Date: _____

Class: _____

ARCADE BUILDER MISSION

Best Booth in the Arcade

You are the new operations manager at NeonPlex Arcade. Two ticket booths are competing for players — Booth A charges \$3 for 5 games, Booth B charges \$5 for 8 games. Players want the best deal, and your job is to crown the winning booth by mastering unit rates.



TODAY'S OBJECTIVES

Content Objective: I can find a unit rate to compare prices and decide the better buy.

Language Objective: I can explain my choice using the words rate, unit rate, per, and better buy.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rates and Unit Rates

Rate

Unit Rate

Per

Ratio

Unit price

Better Buy

Divide



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A rate compares different units; a compares a quantity to exactly 1 unit.

2

A ratio that compares two quantities with different units is a .

3

A rate that tells the cost or amount for one unit is a .

4

The word that means 'for each one' is .

5

A comparison of two quantities is a .

6 A unit rate that tells the cost of one item is the .

7 The option with the lower unit rate, costing less per item, is the .

8 To find a unit rate, I the price by the number of units.

Write About the Math

Which size is the better deal, and how do you know?

¿Qué tamaño es la mejor oferta y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Rates and Unit Rates**
Tasas y tasas unitarias

☐ **Rate**
Tasa

☐ **Unit Rate**
Tasa unitaria

☐ **Per**
Por

☐ **Ratio**
Razón

Model: *It is hard to compare because the booths have different numbers of games.* — Students notice the prices cannot be compared directly because the number of games differs, so they need the cost for one game (the unit rate) to compare fairly.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The ____ bag is the better deal because its unit rate is \$____ per cup, which is ____ than \$____ per cup for the other size.

- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.

Mi afirmación es ____.

- My evidence is ____, so ____.

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Rates and Unit Rates
2. **Notes 2:** Rate
3. **Notes 3:** Unit Rate
4. **Notes 4:** Per
5. **Notes 5:** Ratio
6. **Notes 6:** Unit price
7. **Notes 7:** Better Buy
8. **Notes 8:** Divide
9. **Watch for this mistake:** *A common mistake in Rates and Unit Rates is comparing total costs instead of unit rates: a student sees Pack X at \$4.00 for 10 tokens and Pack Y at \$6.00 for 16 tokens and picks Pack X just because \$4.00 is the smaller number, without dividing to find the cost per token (\$0.40 for Pack X vs. \$0.375 for Pack Y — Pack Y is actually the better buy). A second common mistake is dividing backwards — finding $\text{tokens} \div \text{cost}$ instead of $\text{cost} \div \text{tokens}$ — which turns 'dollars per token' into 'tokens per dollar' and gives an answer that looks reasonable but answers the wrong question. Always divide the total cost by the number of items, and check that your unit rate answers 'the cost for ONE item,' not the reverse.*
10. **Try It 1:** A. Pulse, at 15 points per minute — Nova: $78 \div 6 = 13$ points per minute. Pulse: $75 \div 5 = 15$ points per minute. $15 > 13$, so Pulse is faster.
11. **Try It 2:** A. Machine B, at 21 coins per play — Machine A: $150 \div 5 = 30$ coins per play. Machine B: $168 \div 8 = 21$ coins per play. $21 < 30$, so Machine B is the better value.